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*****
*                                     *
*      STAAD.Pro                     *
*      Version 2003   Bld 1003.ASIA *
*      Proprietary Program of       *
*      Research Engineers, Intl.     *
*      Date=   MAR 3, 2012          *
*      Time=   21:39:40             *
*                                     *
*      USER ID: NCK Associates      *
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INPUT FILE: raft.STD

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1. STAAD SPACE RAFT
2. START JOB INFORMATION
3. ENGINEER DATE 03-MAR-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 1 0 0 0; 2 9.2 0 0; 3 9.2 0 -7.8; 4 -2.4 0 -7.8; 5 -2.4 0 -1.5; 6 0 0 -1.5
9. 7 0.3 0 0; 8 0 0 -0.3; 9 9.2 0 -0.3; 10 0.3 0 -0.3; 11 1.8 0 0; 12 4.1 0 0
10. 13 6.5 0 0; 14 1.8 0 -0.3; 15 4.1 0 -0.3; 16 6.5 0 -0.3; 17 -2.4 0 -1.2
11. 18 0 0 -1.2; 19 -2.1 0 -1.5; 20 -2.4 0 -3; 21 -2.4 0 -7.5; 22 -2.4 0 -4.5
12. 23 -2.4 0 -6; 24 -2.1 0 -7.8; 25 -2.1 0 -3; 26 -2.1 0 -7.5; 27 -2.1 0 -4.5
13. 28 -2.1 0 -6; 29 0.3 0 -1.5; 30 0.3 0 -7.8; 31 0.3 0 -3; 32 0.3 0 -7.5
14. 33 0.3 0 -4.5; 34 0.3 0 -6; 35 1.8 0 -1.5; 36 1.8 0 -7.8; 37 1.8 0 -3
15. 38 1.8 0 -7.5; 39 1.8 0 -4.5; 40 1.8 0 -6; 41 1.8 0 -5.4; 42 4.1 0 -1.5
16. 43 4.1 0 -7.8; 44 4.1 0 -3; 45 4.1 0 -7.5; 46 4.1 0 -4.5; 47 4.1 0 -6
17. 48 4.1 0 -5.4; 49 6.5 0 -1.5; 50 6.5 0 -7.8; 51 6.5 0 -3; 52 6.5 0 -7.5
18. 53 6.5 0 -4.5; 54 6.5 0 -6; 55 6.5 0 -5.4; 56 8.9 0 -1.5; 57 8.9 0 -7.8
19. 58 8.9 0 -3; 59 8.9 0 -7.5; 60 8.9 0 -4.5; 61 8.9 0 -6; 62 8.9 0 -5.4
20. 63 1.05 0 0; 64 1.05 0 -0.3; 65 3.8 0 0; 66 2.8 0 0; 67 2.8 0 -0.3
21. 68 3.8 0 -0.3; 69 6.1 0 0; 70 5.1 0 0; 71 5.1 0 -0.3; 72 6.1 0 -0.3
22. 73 8.5 0 0; 74 7.5 0 0; 75 7.5 0 -0.3; 76 8.5 0 -0.3; 77 -2.1 0 -1.2
23. 78 -0.7 0 -1.2; 79 -1.4 0 -1.2; 80 -0.7 0 -1.5; 81 -1.4 0 -1.5; 82 0.3 0 -1.2
24. 83 -2.4 0 -2.25; 84 -2.1 0 -2.25; 85 0.3 0 -2.25; 86 -1.3 0 -3; 87 -0.5 0 -3
25. 88 -1.3 0 -2.25; 89 -0.5 0 -2.25; 90 -2.4 0 -3.75; 91 -2.4 0 -5.25
26. 92 -2.4 0 -6.75; 93 -2.1 0 -3.75; 94 0.3 0 -3.75; 95 -2.1 0 -5.25
27. 96 0.3 0 -5.25; 97 -2.1 0 -6.75; 98 0.3 0 -6.75; 99 -1.3 0 -3.75
28. 100 -0.5 0 -3.75; 101 -1.3 0 -4.5; 102 -0.5 0 -4.5; 103 -1.3 0 -5.25
29. 104 -0.5 0 -5.25; 105 -1.3 0 -6; 106 -0.5 0 -6; 107 -1.3 0 -6.75
30. 108 -0.5 0 -6.75; 109 -1.3 0 -7.5; 110 -0.5 0 -7.5; 111 -0.5 0 -7.8
31. 112 -1.3 0 -7.8; 113 1.8 0 -0.9; 114 4.1 0 -0.9; 115 6.5 0 -0.9
32. 116 9.2 0 -0.9; 117 8.9 0 -0.9; 118 1.05 0 -1.5; 119 1.05 0 -0.9
33. 120 2.8 0 -0.9; 121 3.8 0 -0.9; 122 2.8 0 -1.5; 123 3.8 0 -1.5; 124 5.1 0 -0.9
34. 125 5.1 0 -1.5; 126 6.1 0 -0.9; 127 6.1 0 -1.5; 128 7.5 0 -0.9; 129 7.5 0 -1.5
35. 130 8.5 0 -0.9; 131 8.5 0 -1.5; 132 9.2 0 -1.5; 133 1.8 0 -2.25
36. 134 4.1 0 -2.25; 135 6.5 0 -2.25; 136 8.9 0 -2.25; 137 2.8 0 -2.25
37. 138 3.8 0 -2.25; 139 2.8 0 -3; 140 3.8 0 -3; 141 1.05 0 -2.25; 142 1.05 0 -3
38. 143 1.05 0 -7.8; 144 1.05 0 -3.75; 145 1.05 0 -4.5; 146 1.05 0 -5.25
39. 147 1.05 0 -6; 148 1.05 0 -6.75; 149 1.05 0 -7.5; 150 1.8 0 -3.75
40. 151 1.8 0 -6.75; 152 2.8 0 -7.8; 153 3.8 0 -7.8; 154 2.8 0 -3.75

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41. 155 3.8 0 -3.75; 156 2.8 0 -4.5; 157 3.8 0 -4.5; 158 2.8 0 -5.4
42. 159 3.8 0 -5.4; 160 2.8 0 -6; 161 3.8 0 -6; 162 2.8 0 -6.75; 163 3.8 0 -6.75
43. 164 2.8 0 -7.5; 165 3.8 0 -7.5; 166 4.1 0 -6.75; 167 4.1 0 -3.75
44. 168 5.1 0 -2.25; 169 5.1 0 -3; 170 6.1 0 -2.25; 171 6.1 0 -3; 172 5.1 0 -7.8
45. 173 6.1 0 -7.8; 174 5.1 0 -3.75; 175 6.1 0 -3.75; 176 5.1 0 -4.5
46. 177 6.1 0 -4.5; 178 5.1 0 -5.4; 179 6.1 0 -5.4; 180 5.1 0 -6; 181 6.1 0 -6
47. 182 5.1 0 -6.75; 183 6.1 0 -6.75; 184 5.1 0 -7.5; 185 6.1 0 -7.5
48. 186 6.5 0 -6.75; 187 6.5 0 -3.75; 188 7.5 0 -2.25; 189 7.5 0 -3
49. 190 8.5 0 -2.25; 191 8.5 0 -3; 192 7.5 0 -7.8; 193 8.5 0 -7.8; 194 7.5 0 -3.75
50. 195 8.5 0 -3.75; 196 7.5 0 -4.5; 197 8.5 0 -4.5; 198 7.5 0 -5.4
51. 199 8.5 0 -5.4; 200 7.5 0 -6; 201 8.5 0 -6; 202 7.5 0 -6.75; 203 8.5 0 -6.75
52. 204 7.5 0 -7.5; 205 8.5 0 -7.5; 206 8.9 0 -6.75; 207 8.9 0 -3.75
53. 208 9.2 0 -2.25; 209 9.2 0 -3; 210 9.2 0 -3.75; 211 9.2 0 -4.5; 212 9.2 0 -5.4
54. 213 9.2 0 -6; 214 9.2 0 -6.75; 215 9.2 0 -7.5; 216 8.9 0 -0.3; 217 8.9 0 0
55. MEMBER INCIDENCES
56. 1 1 7; 2 2 9; 3 3 57; 4 1 8; 6 5 83; 7 7 63; 8 8 18; 9 9 116; 12 11 66
57. 13 12 70; 14 13 74; 21 18 78; 22 5 17; 24 20 90; 25 21 4; 26 22 91; 27 23 92
58. 28 24 4; 34 30 111; 46 36 143; 54 43 153; 62 50 173; 70 57 193; 77 63 11
59. 82 65 12; 83 66 65; 91 69 13; 92 70 69; 100 73 217; 101 74 73; 113 77 17
60. 115 78 79; 116 79 77; 126 83 20; 135 90 22; 136 91 23; 137 92 21; 184 111 112
61. 185 112 24; 206 116 132; 260 132 208; 297 143 30; 330 152 36; 332 153 152
62. 391 172 43; 393 173 172; 458 192 50; 460 193 192; 490 208 209; 492 209 210
63. 494 210 211; 496 211 212; 498 212 213; 500 213 214; 502 214 215; 504 215 3
64. 545 217 2
65. ELEMENT INCIDENCES SHELL
66. 79 1 8 10 7; 80 7 10 64 63; 81 63 64 14 11; 88 11 14 67 66; 89 66 67 68 65
67. 90 65 68 15 12; 97 12 15 71 70; 98 70 71 72 69; 99 69 72 16 13
68. 106 13 16 75 74; 107 74 75 76 73; 119 77 19 81 79; 120 79 81 80 78
69. 121 78 80 6 18; 124 18 6 29 82; 125 8 18 82 10; 150 5 83 84 19
70. 151 19 84 88 81; 152 81 88 89 80; 153 80 89 6; 154 6 89 85 29; 155 83 20 25 84
71. 156 84 25 86 88; 157 88 86 87 89; 158 89 87 31 85; 159 20 90 93 25
72. 160 90 22 27 93; 161 22 91 95 27; 162 91 23 28 95; 163 23 92 97 28
73. 164 92 21 26 97; 165 21 4 24 26; 168 86 99 100 87; 171 99 101 102 100
74. 174 101 103 104 102; 177 103 105 106 104; 180 105 107 108 106
75. 183 107 109 110 108; 186 26 24 112 109; 187 109 112 111 110; 188 110 111 30 32
76. 189 25 93 99 86; 190 93 27 101 99; 191 27 95 103 101; 192 95 28 105 103
77. 193 28 97 107 105; 194 97 26 109 107; 195 87 100 94 31; 196 100 102 33 94
78. 197 102 104 96 33; 198 104 106 34 96; 199 106 108 98 34; 200 108 110 32 98
79. 213 10 82 119 64; 214 64 119 113 14; 215 82 29 118 119; 216 119 118 35 113
80. 241 14 113 120 67; 242 113 35 122 120; 243 67 120 121 68; 244 120 122 123 121
81. 245 68 121 114 15; 246 121 123 42 114; 247 15 114 124 71; 248 114 42 125 124
82. 249 71 124 126 72; 250 124 125 127 126; 251 72 126 115 16; 252 126 127 49 115
83. 253 16 115 128 75; 254 115 49 129 128; 255 75 128 130 76; 256 128 129 131 130
84. 258 130 131 56 117; 262 117 56 132 116; 277 35 133 137 122
85. 278 122 137 138 123; 279 123 138 134 42; 284 133 37 139 137
86. 285 137 139 140 138; 286 138 140 44 134; 293 29 85 141 118; 294 118 141 133 35
87. 295 85 31 142 141; 296 141 142 37 133; 311 31 94 144 142; 312 94 33 145 144
88. 313 33 96 146 145; 314 96 34 147 146; 315 34 98 148 147; 316 98 32 149 148
89. 317 32 30 143 149; 318 17 5 19 77; 323 142 144 150 37; 324 144 145 39 150
90. 325 145 146 41 39; 326 146 147 40 41; 327 147 148 151 40; 328 148 149 38 151
91. 329 149 143 36 38; 358 37 150 154 139; 359 139 154 155 140; 360 150 39 156 154
92. 361 154 156 157 155; 362 39 41 158 156; 363 156 158 159 157; 364 41 40 160 158
93. 365 158 160 161 159; 366 40 151 162 160; 367 160 162 163 161
94. 368 151 38 164 162; 369 162 164 165 163; 370 38 36 152 164
95. 371 164 152 153 165; 372 165 153 43 45; 375 163 165 45 166; 376 161 163 166 47
96. 377 159 161 47 48; 378 157 159 48 46; 381 155 157 46 167; 382 140 155 167 44

97. 423 42 134 168 125; 424 125 168 170 127; 425 127 170 135 49
98. 426 134 44 169 168; 427 168 169 171 170; 428 170 171 51 135
99. 429 44 167 174 169; 430 169 174 175 171; 431 171 175 187 51
100. 432 167 46 176 174; 433 174 176 177 175; 434 175 177 53 187; 435 46 48 178 176
101. 436 176 178 179 177; 437 177 179 55 53; 438 48 47 180 178; 439 178 180 181 179
102. 440 179 181 54 55; 441 47 166 182 180; 442 180 182 183 181; 443 181 183 186 54
103. 444 166 45 184 182; 445 182 184 185 183; 446 183 185 52 186; 447 45 43 172 184
104. 448 184 172 173 185; 449 185 173 50 52; 506 49 135 188 129; 507 135 51 189 188
105. 508 51 187 194 189; 509 187 53 196 194; 510 53 55 198 196; 511 55 54 200 198
106. 512 54 186 202 200; 513 186 52 204 202; 514 52 50 192 204; 515 129 188 190 131
107. 516 188 189 191 190; 517 189 194 195 191; 518 194 196 197 195
108. 519 196 198 199 197; 520 198 200 201 199; 521 200 202 203 201
109. 522 202 204 205 203; 523 204 192 193 205; 524 131 190 136 56
110. 525 56 136 208 132; 526 190 191 58 136; 527 136 58 209 208; 528 191 195 207 58
111. 529 58 207 210 209; 530 195 197 60 207; 531 207 60 211 210; 532 197 199 62 60
112. 533 60 62 212 211; 534 199 201 61 62; 535 62 61 213 212; 536 201 203 206 61
113. 537 61 206 214 213; 538 203 205 59 206; 539 206 59 215 214; 540 205 193 57 59
114. 541 59 57 3 215; 547 76 130 117 216; 548 216 117 116 9; 549 73 76 216 217
115. 550 217 216 9 2
116. DEFINE MATERIAL START
117. ISOTROPIC MATERIAL1
118. E 2.5E+007
119. POISSON 0.17
120. DENSITY 24
121. DAMP 2.8026E-044
122. END DEFINE MATERIAL
123. ELEMENT PROPERTY
124. 79 TO 81 88 TO 90 97 TO 99 106 107 119 TO 121 124 125 150 TO 165 168 171 174 -
125. 177 180 183 186 TO 200 213 TO 216 241 TO 256 258 262 277 TO 279 284 TO 286 -
126. 293 TO 296 311 TO 318 323 TO 329 358 TO 372 375 TO 378 381 382 423 TO 449 -
127. 506 TO 541 547 TO 550 THICKNESS 0.3
128. MEMBER PROPERTY AMERICAN
129. 1 TO 4 6 TO 9 12 TO 14 21 22 24 TO 28 34 46 54 62 70 77 82 83 91 92 100 101 -
130. 113 115 116 126 135 TO 137 184 185 206 260 297 330 332 391 393 458 460 490 -
131. 492 494 496 498 500 502 504 545 PRIS YD 0.3 ZD 0.3
132. CONSTANTS
133. MATERIAL MATERIAL1 MEMB 1 TO 4 6 TO 9 12 TO 14 21 22 24 TO 28 34 46 54 62 -
134. 70 77 79 TO 83 88 TO 92 97 TO 101 106 107 113 115 116 119 TO 121 124 TO 126 -
135. 135 TO 137 150 TO 165 168 171 174 177 180 183 TO 200 206 213 TO 216 -
136. 241 TO 256 258 260 262 277 TO 279 284 TO 286 293 TO 297 311 TO 318 -
137. 323 TO 330 332 358 TO 372 375 TO 378 381 382 391 393 423 TO 449 458 460 490 -
138. 492 494 496 498 500 502 504 506 TO 541 545 547 TO 550
139. SUPPORTS
140. 1 3 TO 217 ELASTIC MAT DIRECT YONLY SUBGRADE 5000
141. 2 FIXED BUT MX MZ KFY 5000
142. LOAD 1 DEAD LOAD
143. SELFWEIGHT Y -1
144. JOINT LOAD
145. 26 FY -2.73
146. 32 FY -7.1
147. 28 FY -3.03
148. 34 FY -10.45
149. 27 FY -3.06
150. 33 FY -10.25
151. 25 FY -3.05
152. 31 FY -10.3

153. 19 FY -2.77
154. 29 FY -7.38
155. 38 FY -17.3
156. 40 FY -12.19
157. 41 FY -9.21
158. 39 FY -15.45
159. 37 FY -16.53
160. 35 FY -24.95
161. 45 FY -29.29
162. 47 FY -7.06
163. 48 FY -4.72
164. 52 FY -29.52
165. 54 FY -7.21
166. 55 FY -4.84
167. 59 FY -16.28
168. 61 FY -11.7
169. 62 FY -8.96
170. 46 FY -3.96
171. 53 FY -4.05
172. 60 FY -15.01
173. 44 FY -9.39
174. 51 FY -9.58
175. 58 FY -15.81
176. 42 FY -33.04
177. 49 FY -32.42
178. 56 FY -21.98
179. 10 FY -1.73
180. 14 FY -8.34
181. 15 FY -9.36
182. 16 FY -7.1
183. LOAD 2 LIVE LOAD
184. JOINT LOAD
185. 26 FY -14.22
186. 32 FY -21.71
187. 28 FY -15.82
188. 34 FY -21.36
189. 27 FY -15.38
190. 33 FY -19.78
191. 25 FY -15.8
192. 31 FY -21.22
193. 19 FY -14.42
194. 29 FY -21.6
195. 38 FY -18.04
196. 40 FY -19.86
197. 41 FY -18.75
198. 39 FY -23.4
199. 37 FY -28.35
200. 35 FY -25.41
201. 45 FY -36.61
202. 47 FY -41.55
203. 48 FY -39.3
204. 52 FY -37.82
205. 54 FY -43.12
206. 55 FY -40.63
207. 59 FY -18.44
208. 61 FY -22.7

209. 62 FY -21.38
210. 46 FY -39.75
211. 53 FY -41.2
212. 60 FY -25.94
213. 44 FY -58.57
214. 51 FY -60.53
215. 58 FY -32.56
216. 42 FY -52.51
217. 49 FY -49.14
218. 56 FY -25.21
219. 10 FY -4.24
220. 14 FY -10.7
221. 15 FY -13.7
222. 16 FY -7.62
223. LOAD COMB 3 1.4DL + 1.6LL
224. 1 1.4 2 1.6
225. LOAD COMB 4 1.0DL + 1.0LL
226. 1 1.0 2 1.0
227. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 217/ 245/ 217
ORIGINAL/FINAL BAND-WIDTH= 215/ 19/ 120 DOF
TOTAL PRIMARY LOAD CASES = 2, TOTAL DEGREES OF FREEDOM = 1302
SIZE OF STIFFNESS MATRIX = 157 DOUBLE KILO-WORDS
REQD/AVAIL. DISK SPACE = 14.8/ 9082.4 MB, EXMEM = 234.0 MB

228. LOAD LIST 4
229. PRINT JOINT DISPLACEMENTS